

NYC TAXI FARE



By

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CONTENTS

1. Introduction.....	3
2. Purpose.....	3
3. Goal.....	3
4. Key Questions	3
5. Metrics and KPIs.....	3
5.1 On-Time Rate.....	3
5.2 Total Rides.....	4
5.3 Average Trip Duration.....	4
5.4 Total Revenue.....	4
6. Data Model.....	4
6.1 Lookup Tables.....	5
6.2 Fact Tables.....	5
6.3 Aggregated Tables.....	5
6.5 Measure Table.....	5
6.6 Relationships.....	5
7. Results.....	6
8. Conclusion.....	8
9. Appendix.....	9

1. Introduction

The NYC taxi fare dataset offers a unique glimpse into how one of the world's busiest cities moves. It's more than just numbers—it's a story of people navigating a vibrant city, shaped by where they go, how much they pay, and how urban dynamics like traffic and congestion impact them. This dataset allows us to explore real-world trends, such as travel patterns, fare structures, and the effect of policies like congestion pricing. It also opens the door to predictive insights that can help improve transportation services, making life easier for both drivers and riders. It's not just about fares; it's about understanding the heartbeat of a city, where millions of journeys intersect every day.

2. Purpose

The purpose of the NYC taxi fare dataset is to provide a detailed and authentic view of urban transportation dynamics in one of the world's busiest cities. By analyzing this data, we aim to uncover insights into travel behavior, fare structures, and the impact of factors such as traffic, time of day, and location on ride costs. This dataset serves as a foundation for understanding city mobility patterns, optimizing transportation services, informing policy decisions, and enhancing the overall commuting experience for millions of people.

3. Goals

The goal of studying the NYC taxi fare dataset is to analyze traffic patterns and congestion periods, track key performance indicators (KPIs), and predict trip times and fare amounts. It also aims to identify key areas for service improvement and fare optimization, enabling data-driven strategies to enhance urban mobility, reduce travel costs, and improve the overall efficiency of taxi services in the city.

4. Key Question

1. What is the most common payment method, and do they affect the tip amount?
2. How do the passenger count and trip distance impact the total fare?
3. What are the busiest days of the week for taxi trips and how do fare amount vary across these days?
4. What are the busiest hours of the day for taxi trips?

5. Metrics and KPIs

A Key Performance Indicator (KPI) is a measurable value used to assess the effectiveness and success of an organization, project, or specific activity in achieving its objectives. KPIs help track progress toward goals by providing quantifiable data points that can be monitored over time. They offer insights into areas that are performing well and those that need improvement, guiding decision-making and strategy.

Here are the key performance indicators (KPIs) focused on in the study, explained in detail:

To measure and analyze the performance and efficiency of taxi services, the following KPIs can be derived from the dataset:

5.1. On-Time Rate: The on-time rate measures the percentage of taxi trips that arrive at the destination within a reasonable or expected time frame. This KPI is essential for assessing the efficiency and reliability of taxi services, helping to identify factors that may cause delays, such as traffic congestion or road conditions. Improving the on-time rate enhances customer satisfaction and ensures a more predictable travel experience.

5.2. Total Rides: This KPI tracks the total number of taxi trips taken over a specific period. It reflects the overall demand for taxi services in NYC and can be used to identify trends in passenger volume, peak travel times, and seasonal variations. Monitoring total rides helps assess the growth or decline in usage, enabling better resource allocation and service planning.

5.3. Average Trip Duration: The average trip duration indicates the typical time taken for a taxi ride from pickup to dropoff. It provides insights into traffic patterns, the impact of congestion, and the efficiency of routes. Tracking this KPI can help identify areas with frequent delays and optimize travel routes to reduce trip times, improving overall service performance.

5.4. Total Revenue: Total revenue measures the overall earnings generated from taxi fares, including base fare, surcharges, tips, and tolls. This KPI is vital for understanding the financial health of the taxi service industry and assessing the effectiveness of fare structures. Analyzing revenue trends can also help identify high-demand areas and times, guiding pricing strategies and service improvements.

6.Data Model

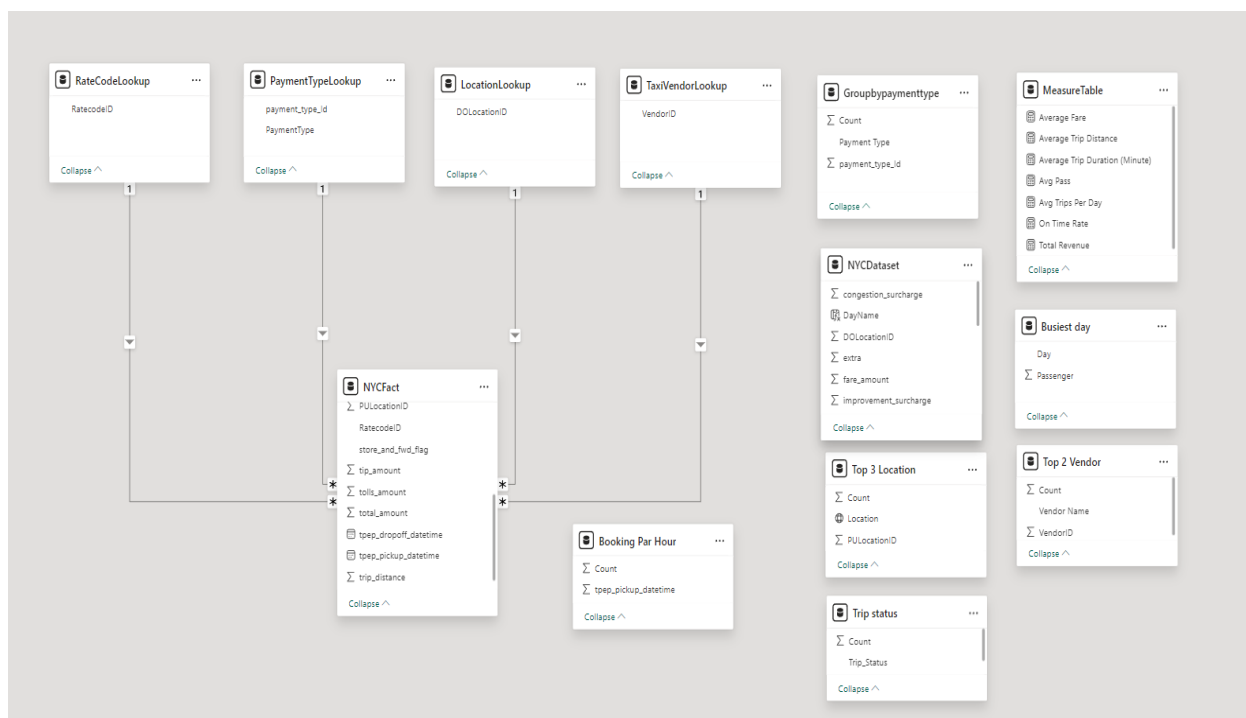


Fig 1. Data Model

The data model in fig 1 represent a data warehouse or data mart for analyzing New York City taxi data. It includes various lookup tables, fact tables, and aggregated data to support analysis of taxi trip metrics, revenue, and patterns. Here's a breakdown of the elements:

6.1 Lookup Tables:

- **RateCodeLookup:** Contains information about different rate codes used in fares, with a RateCodeID.
- **PaymentTypeLookup:** Stores payment types with payment_type_Id and PaymentType descriptions (e.g., cash, card).
- **LocationLookup:** Maps pickup and drop-off locations with identifiers like DOLocationID.
- **TaxiVendorLookup:** Lists taxi vendors using a VendorID.

6.2 Fact Tables:

- **NYCFact:** Central fact table storing detailed trip data such as:
 - PULocationID and RatecodeID for location and fare rate.
 - Financial metrics like tip_amount, tolls_amount, and total_amount.
 - Time-based data (tpep_pickup_datetime and tpep_dropoff_datetime).
 - Trip metrics (trip_distance).

6.3 Aggregated Tables:

- **Groupbypaymenttype:** Aggregates data by payment type, storing the count of occurrences.
- **Booking Par Hour:** Counts bookings per hour based on tpep_pickup_datetime.
- **Busiest Day:** Identifies the day with the most passengers.
- **Top 3 Location:** Lists the top three pickup or drop-off locations based on count.
- **Top 2 Vendor:** Shows the top two vendors based on the count of trips.
- **Trip Status:** Aggregates data by trip status (likely indicating completed, canceled, etc.).

6.4 Measure Table:

- **MeasureTable:** Holds calculated measures such as average fare, average trip distance, average trip duration, average trips per day, on-time rate, and total revenue.

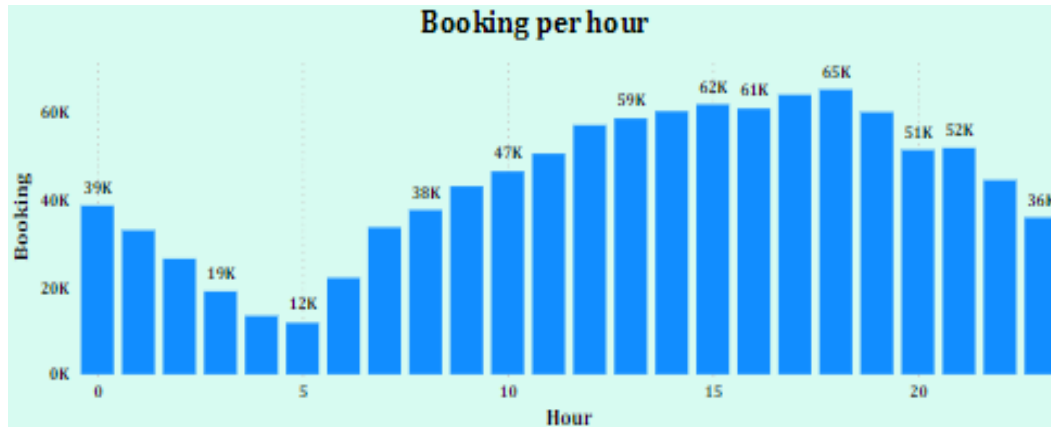
6.5 Relationships:

- Lookup tables are connected to the NYCFact table through keys like RateCodeID, VendorID, and location IDs, allowing for detailed analysis and filtering.
- Aggregated tables connect to fact tables for summarizing and visualizing data trends.

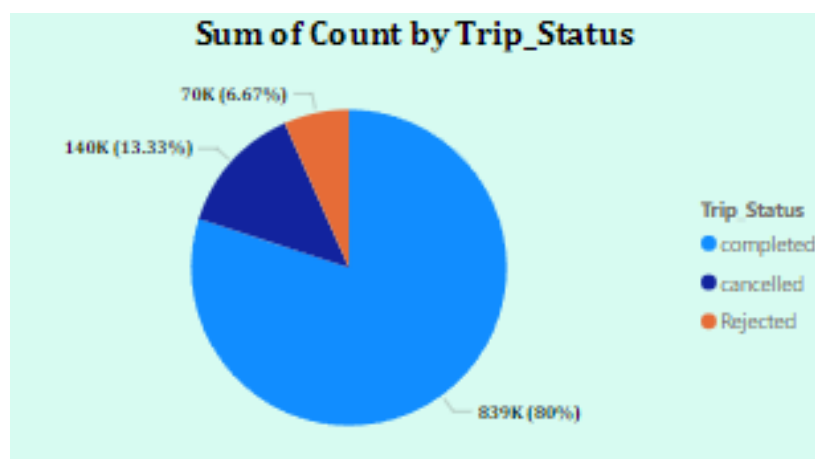
Overall, the model supports analyzing taxi trips in terms of revenue, patterns, time-based trends, and vendor performance.

7. Results

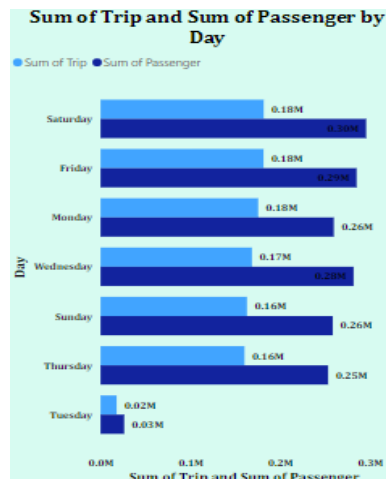
The bar graph titled "Booking per hour" shows the number of bookings (trips) for each hour of the day. The x-axis represents the hours (0 to 23), and the y-axis represents the number of bookings. The data suggests that bookings start relatively low during the early morning hours (around 12 AM to 6 AM), increase steadily from 7 AM, and peak around the late afternoon (3 PM to 5 PM). Afterward, they gradually decrease towards the night



Pie Chart: "Sum of Count by Trip_Status. This chart categorizes the trips based on their status: "completed," "cancelled," and "rejected." "Completed" trips make up the majority (80%, or 839K), showing that most trips reached their destination. "Cancelled" trips account for 13.33% (140K), while "rejected" trips make up 6.67% (70K), indicating a significant portion of trips did not go through as planned.



Bar Chart: "Sum of Trip and Sum of Passenger by Day. This chart displays the total number of trips and passengers for each day of the week. The bars represent two metrics: "Sum of Trip" (light blue) and "Sum of Passenger" (dark blue), showing how they vary across different days. The data indicates that Wednesday has the highest number of trips (0.18M) and passengers (0.26M), suggesting higher demand on that day. Tuesday has the lowest numbers for both trips (0.02M) and passengers (0.03M), indicating less taxi activity.



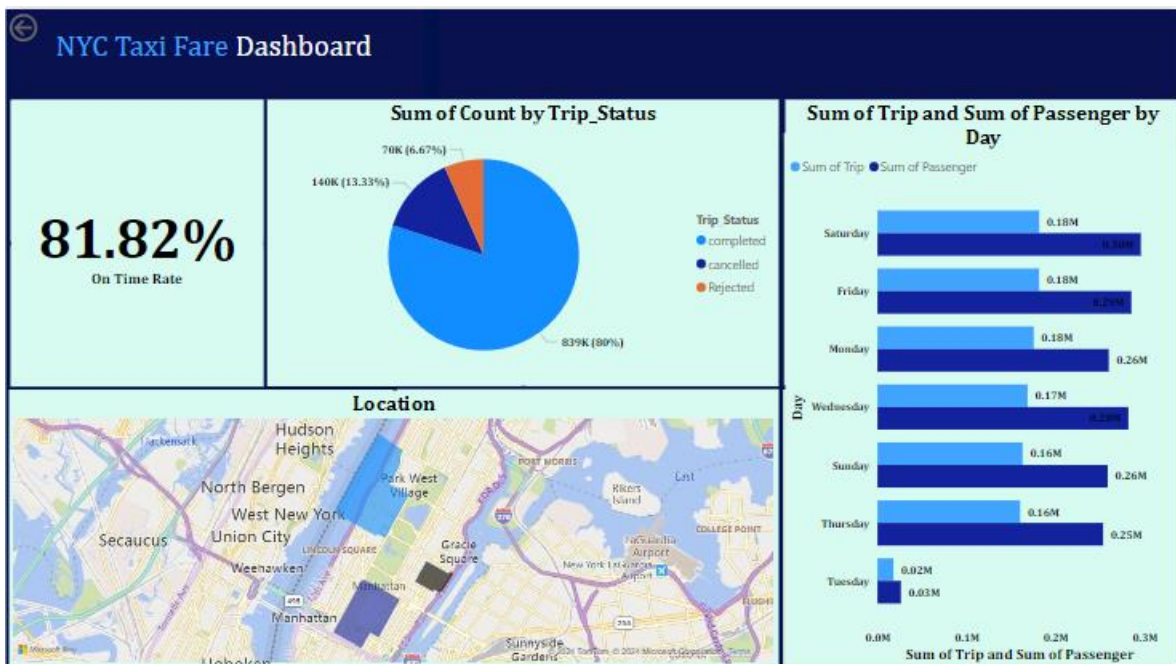
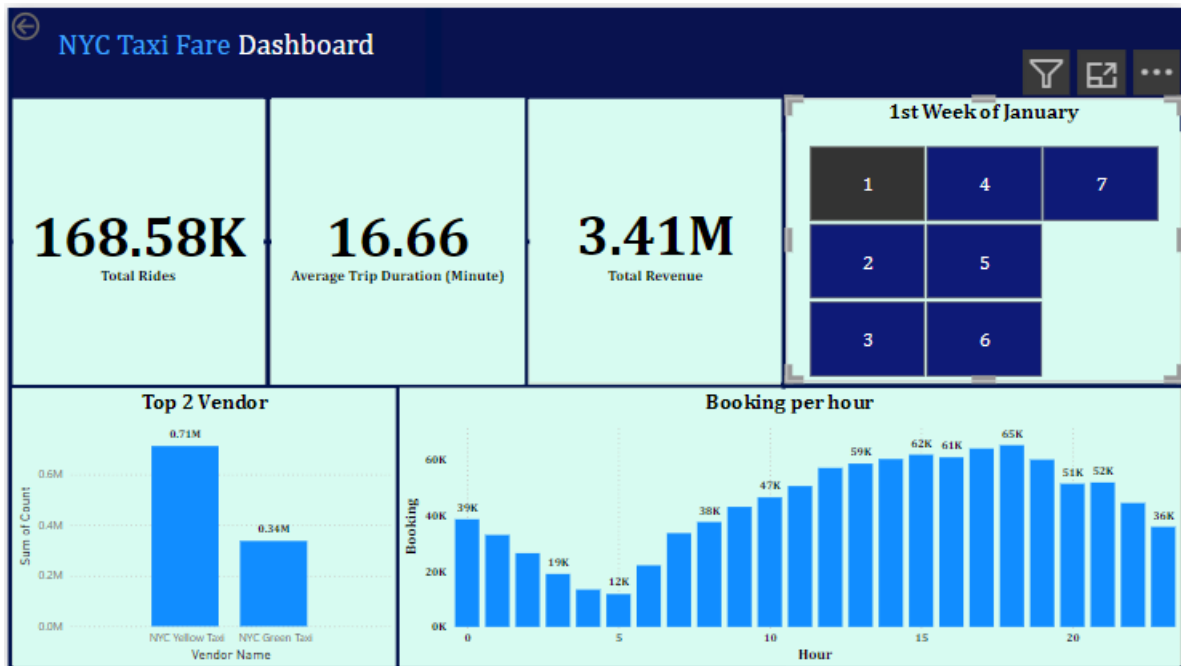
The map in the image displays parts of New York City including locations such as Manhattan, Hudson Heights, West New York, and North Bergen. Highlighted Areas represent zones where taxi pickups or drop-offs frequently occur. For example, the areas around Manhattan, including Midtown, might have higher activity due to tourist attractions, business centers, and major transit hubs. By mapping these locations, the dashboard can provide insights into which regions have the most taxi activity, peak times for pickups/drop-offs, or how trip patterns vary geographically across New York City.



8.Conclusions

1. Demand Patterns: The data indicates high demand for taxi services during late-night hours and special occasions. Taxi companies can use this information to optimize fleet deployment and ensure availability during peak times.
2. Fare Structure: The variability in fare amounts suggests that the fare structure is influenced by multiple factors, including distance, time, and additional charges. Understanding these factors can help in setting fair and competitive pricing strategies.
3. Payment Preferences: The preference for credit card payments suggests that enhancing electronic payment options could improve customer satisfaction and streamline operations.
4. Service Optimization: By identifying common pickup and dropoff locations, taxi services can optimize routes and reduce wait times for passengers. This can lead to improved efficiency and customer experience.
5. Policy Implications: The impact of congestion surcharges highlights the need for policies that manage traffic flow and reduce congestion in busy areas. This can improve travel times and reduce environmental impact.
6. Passenger Insights: Understanding passenger behavior, such as the typical number of passengers per trip, can help in designing services that cater to the needs of different customer segments

9.Appendix: Dashboard



Link for interactive dashboard

<https://app.powerbi.com/groups/me/reports/7e01d608-0218-4314-9749-3f410f0aad67/b075d42b182273ae0083?experience=power-bi>

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